



Perfluorocarbon-loaded micro and nanosystems for medical imaging: A state of the art



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ABSTRACT

Perfluorocarbons (PFCs) are fluorinated compounds widely used in the past as oxygen carriers. They have been the focus of research as contrast agents for ultrasonography and fluorine magnetic resonance imaging (MRI). For ultrasonography, most contrast agents use gaseous PFCs and some are approved for clinical use. Liquid PFCs, however, are being studied for the past decade due to their improved stability under ultrasound waves. For fluorine MRI, liquid PFCs are preferred due to their higher fluorine concentration, but for the moment no product is on the market yet. For both these imaging modalities, a large body of formulation is needed to allow safe PFC injection in the bloodstream. In this review, we will focus on PFC encapsulation into micro and nanosystems for medical imaging purposes, from research to the clinics.

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1. Introduction

Perfluorocarbons (PFCs) are organic molecules in which all hydrogen atoms in the molecule are replaced by fluorine. Due to the specificity of the carbon–fluorine bond, their physico-chemical properties render them attractive for medical applications. High oxygen solubility, low surface tension, hydrophobicity and lipophobicity, inertness and absence of metabolism are their most interesting characteristics for clinical applications [1].

Depending of their molecular weight and chemical structure, PFCs may exist as gases, as liquids or even solids (Table 1). Although gaseous PFCs present some toxicity, liquid PFCs are relatively safe [2]. Liquid PFCs were initially mainly used as artificial blood substitutes for liquid ventilation due to their high oxygen solubility and therefore were extensively studied in terms of biocompatibility and toxicity [3–6]. PFCs have been also tested as contrast agents for ultrasonography and fluorine magnetic resonance imaging (MRI) since the end of the 1970s [2]. On one hand, for ultrasonography, most contrast agents use gaseous PFCs,

although liquid PFCs are being studied for the past 15 years [7–14]. On the other hand, for fluorine MRI, liquid PFCs are preferred since they lead to higher fluorine concentration [15–17]. For both these imaging modalities, a large body of formulation is needed to allow safe PFC injection in the bloodstream. Indeed, one cannot inject a gas intravenously without risking embolism. Formulation is therefore needed to produce stable and calibrated bubbles that would transit unimpeded through the pulmonary capillaries. For liquid PFCs, given their poor solubility in water, they have been formulated either as nanoemulsions [18] or nanocapsules [17] to allow intravenous injection. In this review, we will focus on PFC encapsulation into micro and nanosystems for medical imaging purposes.

2. Perfluorocarbons for ultrasonography

Ultrasonography or ultrasound imaging is based on acoustic signals produced by reflection or backscattering of acoustic waves above the audible frequency range for humans (>20 kHz). This technique provides important information on the anatomy and function of internal organs or on the blood flow and is one of the most commonly performed non-invasive diagnostic tests used today [19]. Unfortunately, it cannot be used to monitor tissue microvasculature given the small size of the microvessels and the poor difference of signal between the blood and the surrounding

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Table 1
Physico-chemical parameters of commonly used PFCs.

PFCs	Formula	MW ^a	Density (g/mL)	BP (°C) ^b
Sulfur hexafluoride	SF ₆	146	NA	−64
Perfluoropropane	C ₃ F ₈	188	NA	−37
Perfluorobutane	C ₄ F ₁₀	238	NA	−2
Perfluoropentane	C ₅ F ₁₂	288	1.62	26–36
Perfluorohexane	C ₆ F ₁₄	338	1.76	58–60
Perfluorodichlorooctane	C ₈ F ₁₆ Cl ₂	471	1.84	115
Perfluorooctyl bromide	C ₈ F ₁₇ Br	499	1.93	140.5
Perfluorononane	C ₉ F ₂₀	488	1.80	126
Perfluorodecalin	C ₁₀ F ₁₈	462	1.99	144
Perfluoro-15-crown-5-ether	C ₁₀ F ₂₀ O ₅	580	1.78	145
Perfluorotributylamine	C ₁₂ F ₂₇ N	671	1.88	173–181
PERFECTA	C ₂₁ H ₈ F ₃₆ O ₄	1008	2.09	ND

^a Molecular weight.

^b Boiling point.

tissue. The use of microbubble contrast agents allows to overcome this limitation. Indeed, due to their micron-size and the gas they contain, ultrasound waves make them oscillate therefore enhancing the backscattered signal. At low acoustic power, they oscillate at the same frequency they are excited (fundamental) whereas at higher power they can oscillate at the double of the excitation frequency (2nd harmonic) or even burst therefore generating signal at subharmonic and harmonic frequencies. The bursting provides the generation of strong acoustic signals that can greatly exceed conventional ultrasound backscattering produced by reflection due to differences in acoustic impedance (Fig. 1) [20].

Gramiak and Shah described the effects of intracardiac injection of agitated saline on the improved aortic root delineation as a consequence of the backscattered ultrasound signal arising from air microbubbles present in the solution [21]. Subsequently, other types of microbubbles were generated (hydrogen peroxide, dextrose, etc) but none of them could pass through the pulmonary and cardiac capillaries due to their excessive sizes. Although sonication was used to favor a significant decrease of their mean diameter, air solubility in the blood led to a rapid disappearance of bubbles from the blood stream few seconds after the administration, compromising their clinical use [20]. The first generation of modern microbubbles, based on the air-loaded biocompatible capsules, showed the same drawback; the first marketed agent, named Albunex[®], made up of 5% human serum albumin, was approved for human use by the U.S.A. Food and Drug Administration (FDA) in 1994. Although the diameter of the systems was <10 μm, the thin thickness of protein shell (30–50 nm) did not prevent the diffusion of entrapped air in the

blood, thus disappearing rapidly after the systemic administration of the formulation (half-life <1 min) and focusing its application only to the left ventricular chamber [22]. Several compounds were investigating to further stabilize air microbubbles with no success [23]. In 2001 Mayer and Grayburn provided the guidelines for a useful transpulmonary contrast agent that “Firstly,” should be given intravenously for ease of administration. The microbubbles must be small enough and of sufficient concentration to traverse the pulmonary circulation and opacify the left heart. Furthermore, the contrast agent should possess a favorable safety profile with little or no biological effects. The agent must be stable and persist in the microcirculation to enable acquisition of ultrasound images of the left ventricle in multiple views. Also, the microbubbles should be confined to the microvasculature without diffusion across the endothelium. This property allows the microbubbles to act as blood pool markers with a behavior similar to that of red blood cells. Lastly, the microbubble should be acoustically responsive” [24]. Researcher efforts have been focusing in fulfilling these requirements.

To formulate more stable microbubbles, air was replaced by PFCs which present a lower water solubility (Table 1). The physico-chemical composition of the formulation influences the stiffness of the bubbles, their resistance to rupture in the ultrasound pressure field and their clearance by the reticuloendothelial system (RES). Different investigations showed that bubbles with a diameter of several micrometers may furnish the most efficient acoustic response while smaller bubbles provide acoustic backscatter and the possibility to extravasate into the tumor tissues [25]. Among gaseous PFCs, sulphur hexafluoride (SF₆) has been stabilized with lipids and marketed under the name Sonovue[®] by Bracco Diagnostics [26]. However, several cases of adverse reactions were reported with Sonovue, mostly probably due to allergic reaction to sulphur [27,28]. Sonovue[®] mean bubble diameter is 2.5 μm. Following intravenous administration of 0.3 mL/kg of SonoVue(-trade mark) (*i.e.*, approximately ten times the imaging dose) in human volunteers, a distribution half-life of about 1 min and an elimination half-life of about 6 min were observed [29] (Table 2).

Biocompatible lipids were used to formulate an IV injectable formulation of perfluoropropane commercialized as Definity[®] (Lantheus Medical Imaging[®], www.definityimaging.com). The formulation consists of microspheres (mean diameter of 1–3 μm) made up of hexadecanoic acid, 1-[(phosphonoxy)methyl]-1,2-ethanediyl ester, monosodium salt (DPPA), (R)-4-hydroxy-*N,N,N*-trimethyl-10-oxo-7-[(1-oxohexadecyl)oxy]-3,4,9-trioxa-4-phosphapentacosan-1-aminium, 4-oxide, inner salt (DPPE) and (R)-α-[6-hydroxy-6-oxido-9-[(1-oxohexadecyl)oxy]-5,7,11-trioxa-2-aza-6-phosphahexacos-1-yl]-ω-methoxypoly(ox-1,2-ethanediyl), monosodium

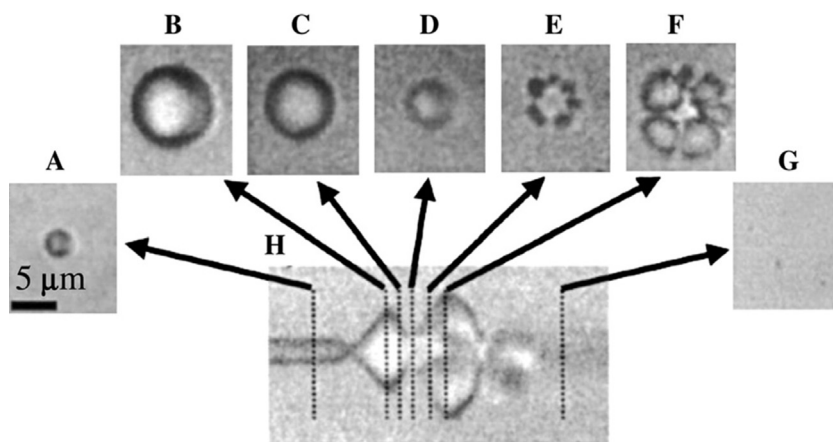


Fig. 1. Optical frame images and streak image corresponding to the oscillation and fragmentation of a contrast agent microbubble, where fragmentation occurs during decompression. The bubble has an initial diameter of 3 μm, shown in (A). The streak image in h shows the diameter of the bubble as a function of time, and dashed lines indicate the times at which the two-dimensional frame images in (A)–(G) were acquired relative to the streak image (from Ref. [20]).

Table 2

Main features of the most important microbubble formulations used for ultrasonography or ultrasound imaging.

Name	Gas	Type/composition	Clinical indications	Approval status	Developer
Albunex [®]	Air	5% human albumin microparticles	Contrast sonography and Doppler-echocardiography	FDA-approved—no longer in production	Mallinckrodt Inc.
Sonovue [®]	Sulfurhexafluoride (SF ₆)	Microbubbles—Macrogol 4000, DSPC, DPPG, Palmitic acid	Not-cardiac imaging—imaging of blood vessels, breast, liver	Available in US and EU	Bracco Diagnostics, Inc.
Definity [®]	Octafluoropropane (C ₃ F ₈)	Microbubbles—DPPA, DPPC, MPEG5000 DPPE	Echocardiography—indicated for use in patients with suboptimal echocardiograms to opacify the left ventricular chamber and to improve the delineation of the left ventricular endocardial border	Available in US and EU	Bristol-Myers Squibb Medical Imaging Inc.
Optison [®]	Octafluoropropane (C ₃ F ₈)	Microbubbles—Albumin <i>N</i> -acetyltryptophan, caprylic acid	Ultrasound—imaging the endocardial borders of the heart in order to see cardiac wall motion abnormalities and to guide the selection and monitoring of treatment	Available in US and EU	Molecular Biosystems Inc.
Imagify [™]	Perflubutane (C ₄ F ₁₀)	PLGA microspheres	Detection of coronary artery disease	FDA and EMA not approved	Acusphere Inc.

salt (MPEG5000 DPPE) containing perfluoropropane, Definity[®] has been approved by the FDA for the clinical application and it is used in “patients with suboptimal echocardiograms to opacify the left ventricular chamber and to improve the delineation of the left ventricular endocardial border”. Fatal cardiopulmonary and anaphylactic events and other serious but nonfatal events have been observed 30 min after the administration even if they are uncommon, while numerous other investigations related to the potential carcinogenesis, mutagenesis, and impairment of fertility are in progress [30]. Perfluoropropane microbubbles were also stabilized by albumin (Optison[®], GE Healthcare) leading to microbubbles of about 4.7 μm of median diameter [10]. Despite these modifications, the microbubble half-life remained short and researchers have considered using polymer to encapsulate microbubbles and extend their half-life.

2.1. PLGA microsystems for gaseous PFC delivery

Although they are characterized by a more rigid and less compressible structure by comparison with lipid/surfactant systems that could lessen PFC dilatation and backscattered signal, it was demonstrated that polymeric microbubbles possess a better *in vivo* stability even after long ultrasound exposure [31].

In particular, polyesters such as polylactide (PLA), polyglycolide (PGA) and poly-D,L-lactide-co-glycolide (PLGA) were used as a consequence of their favorable technological properties. These polymers are broadly used in the pharmaceutical field and approved by the FDA for human use [32]. PLA/PGA/PLGA are normally metabolized into water and carbon dioxide by the tricarboxylic acid cycle, thus avoiding any accumulation phenomena and related side effect and are therefore used in drug delivery as an ideally safe material. Micro- and nanoparticles containing active compounds (*i.e.* antioxidant, antitumor, antiviral drugs) with different physico-chemical properties were formulated to improve drug biopharmaceutical properties [33].

A typical example of polyester-based ultrasound contrast agent is AI-700 (Imagify[™], Acusphere Inc., U.S.A.). It consists of a lyophilized preparation of porous, synthetic, biodegradable PLGA microspheres containing perfluorobutane gas [34]. Besides PLGA, the formulation contains mannitol to maintain the osmolarity at 300 mOsm/L, polysorbate 80 and the 1,2-diarachidoyl-*sn*-glycero-3-phosphocholine (DAPC) phospholipid useful to regulate the gas leakage from the polymeric matrix and to prevent particle aggregation. AI-700 possesses a diameter of about 2.7 μm , a negative Zeta-potential

(~ -50 mV) and a high stability after exposure to high-power clinical ultrasound. In 2005, Straub and co-workers modulated the size of pores by using different solutions of ammonium bicarbonate (Fig. 2) and demonstrated that the *in vitro* acoustic backscattering from AI-700 at 2.25 MHz was stable for more than 10 min as compared with the signal backscattered from Optison[™] (perfluoropropane-loaded human albumin) that rapidly decreased in 2 min [34].

AI-700 was also investigated in terms of pharmacokinetics and biodistribution in rats after intravenous administration [35] with a specific focus on the exhaled elimination kinetics of perfluorobutane. Microspheres labeling with palladium (Pd) was developed to indirectly quantify PLGA by measuring Pd by inductively coupled plasma-mass spectrometry (ICP-MS). Pd labeling did not modify PLGA microspheres physico-chemical properties and most of the metal remained associated to the polymer with only $\sim 10\%$ Pd released after 24 h. The quantification of Pd in the bloodstream after *i.v.* injection of AI-700 (0.97 mg Pd/kg) in Sprague–Dawley rats clearly evidenced that the C_{max} was reached 1–2 min after injection, while the Pd blood concentration decreased up to 85% 5 min after the maximum plasmatic concentration. Moreover, the formulation was highly localized in the lungs, liver and spleen (50–70% of the initial dose) 24 h after administration. Interestingly, it was evidenced that most of perfluorobutane was expired within the first 10 min (60%) even if small traces of the gas were detected 48 h after administration, showing a noticeable increase of the residence time with respect to other bubble formulations like Optison[™] and Definity[™] [35].

The increased stability of AI-700 offers the opportunity of longer visualization times to assess both myocardial perfusion and wall motion and diagnose coronary artery disease. For example, it was demonstrated that perfusion stress echocardiography (PSE) with AI-700 is clinically equivalent to photon emission computed tomography (SPECT) for the detection of coronary artery disease in stable chest pain patients being evaluated for inducible ischemia (Fig. 3), thus allowing to avoid radiation use and lowering costs [36]. The most common side effects are headache, chest pain, nausea, flushing, and chest discomfort. At the end of phase III studies, Acusphere has submitted the Marketing Authorization Application (MAA) for Imagify[™] to the European Medicines Agency (EMA) in the end of 2012, while it has reached agreement with the FDA under a Special Protocol Assessment on the design of a placebo trial to demonstrate that stress ultrasound with Imagify[™] had superior efficacy (www.acusphere.com).

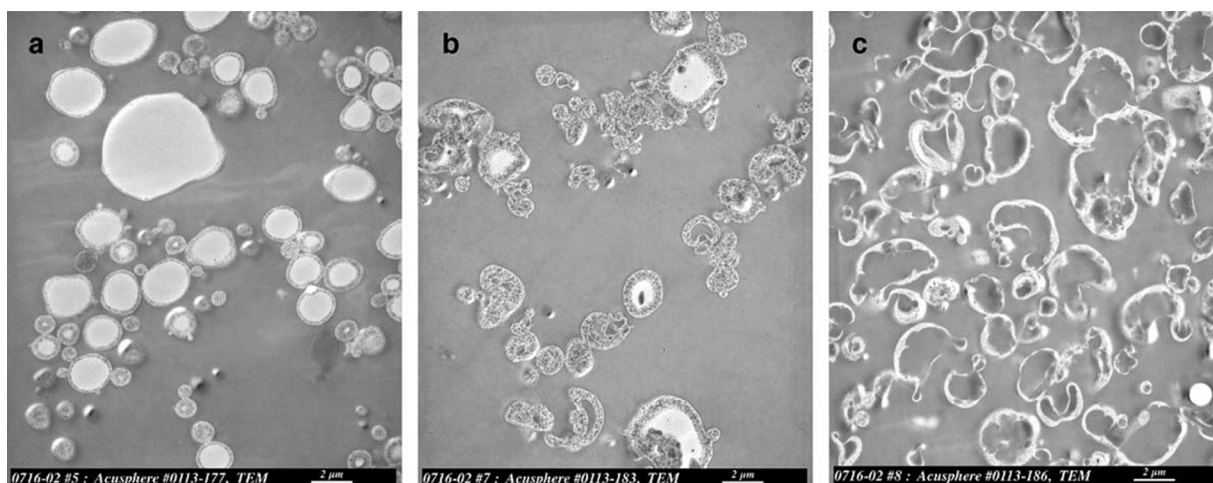


Fig. 2. TEM images of the cross-sections of representative PLGA microparticles. The ratio of ammonium bicarbonate containing solution:PLGA containing solution was (a) 0:100, (b) 2:98, (c) 9:91 (from Ref. [34]).

2.2. PLGA microsystems for liquid PFC delivery

PLGA was the main component of other formulations containing liquid PFCs proposed as ultrasonic contrast agents [14,37,38]. In particular, different PFCs with different boiling points: perfluorooctyl bromide (PFOB), perfluoropentane, perfluorohexane (PFH), perfluorooctane and perfluorodecalin, have been encapsulated into polymeric capsules by a modified emulsion–evaporation process (Fig. 4).

PFCs are added to the organic phase containing the polymer in such way that all components are miscible. The organic phase is then emulsified with an aqueous phase containing a surfactant. As the solvent evaporates, the PFC nucleates in the organic droplet. Capsules are then formed as the polymer precipitates around the PFC droplet. It was shown that both the choice of the surfactant and the polymer are crucial for regular capsule formation. For example, several surfactants were reviewed to ensure the formation of capsules with a polymer shell of PLGA and a liquid core of PFOB [39]: polyvinyl alcohol (PVA), sodium cholate (SC) and sodium taurocholate (TC). PVA, the most used pharmaceutical surfactant for the formulation of polymeric micro/nanosystems [40] induced the appearance of “acorn” objects made up of a liquid hemispherical PFOB droplet and a solid PLGA hemisphere along with regular capsules (Fig. 5). It was demonstrated that the use of TC (1.5% w/v) led to the formation of only “acorn” objects, while SC at the same concentration induced the formation of regular capsules with a core–shell morphology (Fig. 5). The polyester end-group can influence capsule morphology as it was observed for PFH encapsulation [38]. This can be explained by a subtle equilibrium of interfacial tensions during solvent evaporation. The core–shell structure is the result of a significant adsorption of the polymer at the dichloromethane–water interface together with a good stability of the PFC droplet within the emulsion globule which signature is a low interfacial tension at the PFC/organic phase interface.

Sodium cholate (SC) seems to be the best surfactant to yield both microcapsules of PFOB, PFH and perfluorodecalin [37–39]. Capsules could not be obtained with perfluoropentane probably due to its low boiling point (36 °C, lower than the boiling point of methylene chloride used in the organic phase). Recently, SC has been used by Kirsch and co-workers to obtain PLGA micro/nanocapsules containing perfluorodecalin. To avoid using of PVA as a cryoprotectant they proposed to replace it by Tetronic-908 to prevent particle aggregation after reconstitution of the lyophilisate [41].

Capsule size can easily be varied by modifying the energy input during the emulsification step (Fig. 4) and using ultraturrax emulsification one can obtain microcapsules while using sonication emulsification one can obtain submicronic capsules. Another interesting feature of these PLGA capsules is their thickness versatility: for both microcapsules and nanocapsules the thickness-to-radius ratio (T/R) depends only on the polymer/PFC proportions. For PLGA-PFOB micro/nanocapsules T/R could be tuned between 0.25 and 1 (spheres) [37] while for PLGA-PFH T/R could be modulated between 0.1 and 1 [38]. Both capsule size and T/R were shown to be important parameters for capsule capacity to backscatter ultrasound waves defined as their echogenicity [14]. As for microbubbles, the larger the capsules the more important the echogenicity [14,42]. For nanocapsules, the thinner the capsules, the more echogenic [14]. Although a lot of models exist to explain thin shell microbubbles ultrasound interactions [43,44], only a few have been developed for polymer capsules containing liquid PFCs [45,46].

These polyester nanocapsules were injected intravenously to mice and allowed to highlight the hepatic vena cava using tissue harmonic imaging (Fig. 6). Although rather intense, the signal disappeared quickly due to dilution of the suspension and to capsule uptake by macrophages of the reticuloendothelial system [14].

Indeed, the systemic administration of colloidal devices requires their coating with specific compounds to prevent their opsonization followed by macrophage uptake in reticuloendothelial organs, to ensure long circulation in the blood compartment [47,48]. To increase the plasmatic half life of colloids, different strategies have been considered including surface modification either with polysaccharides (dextran, heparin, chitosan, hyaluronan) [49–51] or polyethylene glycol (PEG) chains [52]. If the density of PEG is high enough, PEG chains adopt a brush conformation preventing opsonization by steric repulsion of blood proteins. PEG is FDA approved and easily eliminated by renal filtration provided its molecular weight is lower than 30 kDa [53]. Although in the last years it was demonstrated that multiple systemic administrations of PEGylated nanosystems favor the accelerated blood clearance (ABC) phenomenon, as a consequence of IgM production, this polymer still represents the best choice for protection from blood proteins adsorption and decreased uptake by the reticuloendothelial system [54]. For example, PEGylated liposomal doxorubicin (Doxil, Caelyx) represents the standard treatment of different cancers in clinical practice, thanks to their

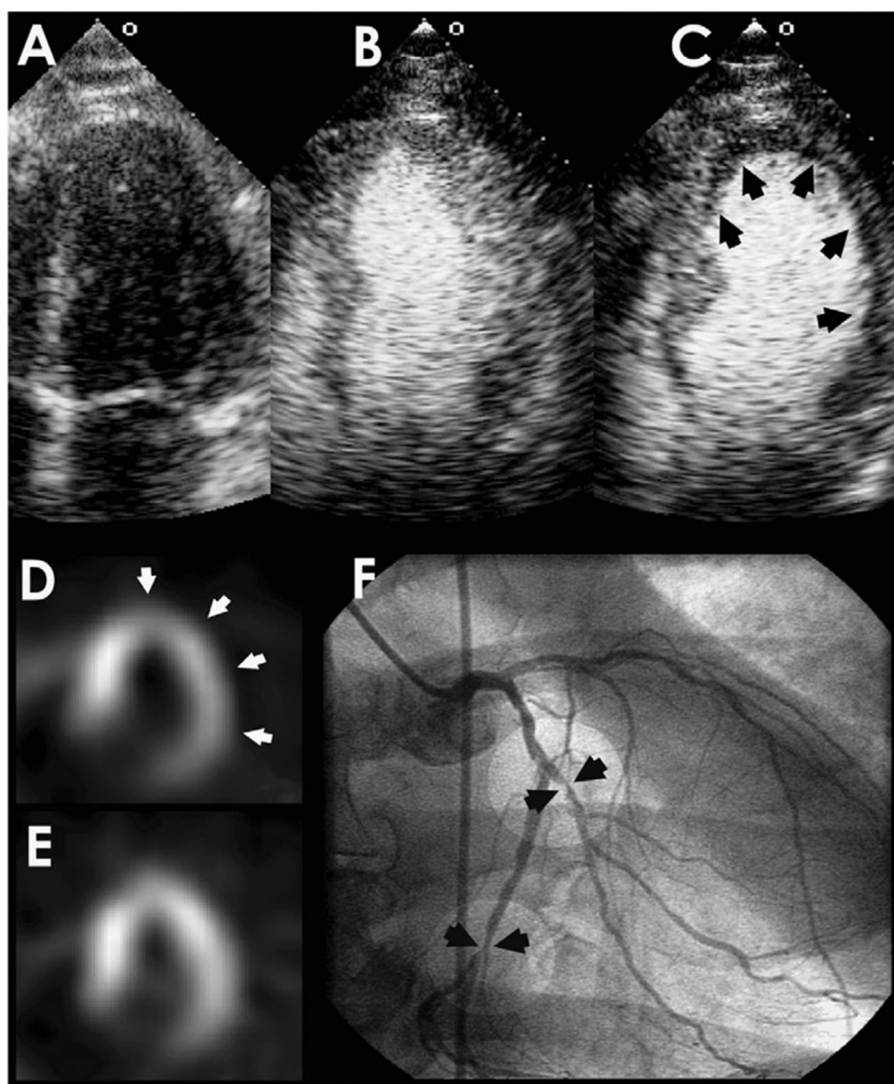


Fig. 3. Detection of coronary artery disease by PSE. Apical 4-chamber view without contrast and after administration of Imagify at rest (A and B). Apical lateral (inducible subendocardial) perfusion defect during PSE (C), corresponding to similar perfusion defect on horizontal long-axis view in stress SPECT (D). SPECT image at rest (E). Angiography with lesions in left circumflex system (F). Images were obtained from the same patient at rest and with dipyridamole. Black arrows indicate subendocardium (C). Arrows indicate perfusion defects (C and D) and angiographic lesions (F) (from Ref. [36]).

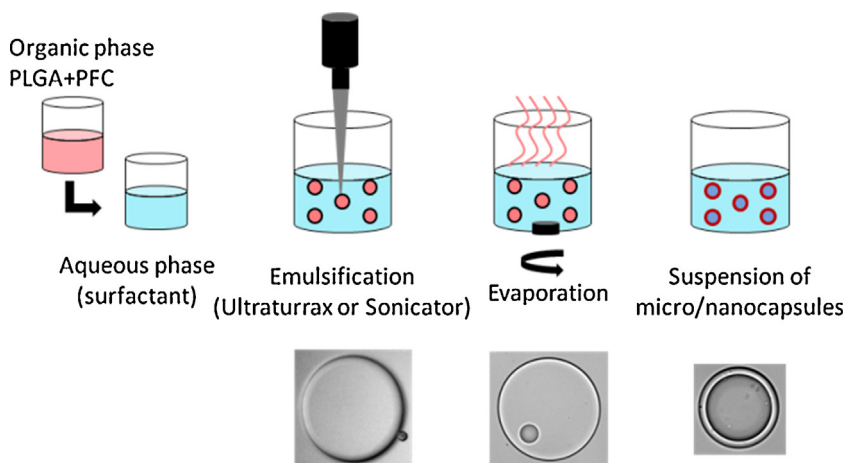


Fig. 4. Preparation of PLGA micro/nanocapsules of PFC by a modified emulsion–evaporation process. Bright field images show the evaporation process leading to the formation of a microcapsule [37,38].

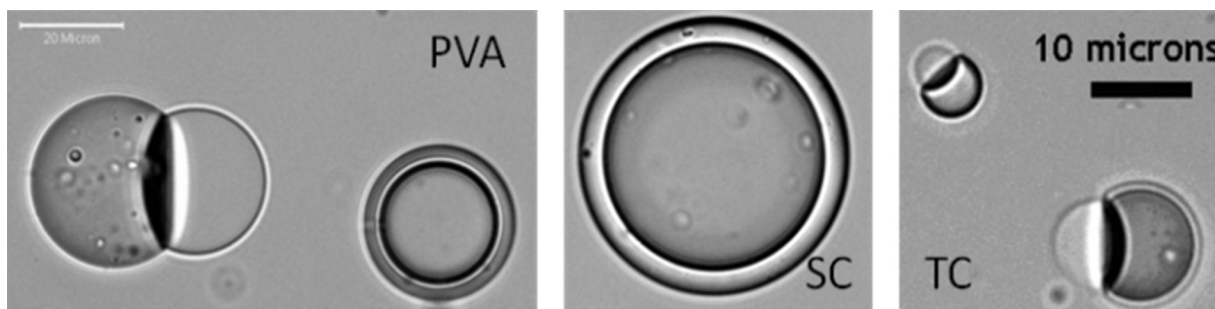


Fig. 5. Influence of the surfactant used on the encapsulation of PFOB into PLGA microobjects [39].

ability to efficiently deliver the antitumor compound in solid tumors through the Enhanced Permeation and Retention (EPR) effect, by increasing the drug plasmatic half-life and decreasing side effects [55–57]. To benefit from the EPR effect and allow PFC capsules to serve as tumor diagnostic agents using ultrasound imaging, two strategies were considered to efficiently coat PLGA/PFOB micro/nanocapsules with PEG.

The first one consisted in adding a PEGylated phospholipid, DSPE-mPEG2000, in the organic phase during the preparation to the interdigitation of phospholipids in the PLGA shell [58,59]. Since capsule mechanism of formation is exquisitely sensitive to interfacial phenomena it was difficult to yield a good PEG coverage, using this strategy. The best PEG coverage obtained by this method expressed as the minimum surface occupied by one

PEG chain, was 100 nm^2 . As PEG chains have a large range of motion, they may adopt a mushroom configuration or leave free spaces between each other allowing blood protein adsorption. In any case, the phospholipid strategy revealed unfavorable to reduce uptake by the RES.

The second strategy for PEG coverage consisted in replacing PLGA by PLGA-PEG5000 copolymer (PLA/PGA 50/50, with 10% w/w of PEG5000). Given the amphiphilic properties of the copolymer nanoprecipitation and emulsion–evaporation methods were considered to yield nanocapsules of PFOB. The emulsion-based technique allowed to obtain nanocapsules with ideal mean sizes ($\sim 120 \text{ nm}$), low polydispersity index (< 0.2) and high encapsulation amount of PFOB as compared with nanoprecipitation [17]. The PEG coating was dense enough to prevent complement activation. The direct injection of PEGylated PFOB capsules into a CT-26 xenograft tumor in mice provided ultrasound signal using a Contrast Pulse Sequencing mode, proving PEGylation did not alter echogenicity [60]. PEGylated nanocapsules were further injected intravenously to mice bearing CT-26 xenograft tumors. Although acquisition time (0 h, 4 h and 7 h), frequencies (7 and 14 MHz), and mechanical indices (0.6–1.9) were modulated, no significant contrast enhancement was observed in the tumor. PEGylated nanocapsules may not be echogenic enough to be detected at low local concentration (up to 4% injected dose in the tumor) by ultrasonography, thus their design need to be further improved [60].

Some authors have combined imaging properties of PFCs with drug loading capacity of polymeric nanosystems to yield theranostic systems. Rapoport and Gao [61,62] developed multifunctional nanoparticles to combine ultrasonic tumor imaging and targeted chemotherapy, using perfluoropentane (PFP) as liquid PFC. Mixtures of drug-loaded polymeric micelles and PFP nanobubbles of poly(ethylene glycol)-*block*-poly(L-lactide) (PEG-PLLA) or PEG-*block*-poly(caprolactone) (PEG-PCL) were prepared and loaded with doxorubicin. At body temperature, nanoparticles are converted into nano/microbubbles, due to the low boiling point of PFP (Table 1). Upon intravenous injection to mice, doxorubicin loaded micelles and nanobubbles accumulated selectively into the tumor interstitium thanks to the EPR effect [63,64] and nanobubbles coalesced into larger, highly echogenic microbubbles. These microbubbles provided ultrasound signal and led to local release of the encapsulated drug triggered by ultrasound [61,62].

As a conclusion, for ultrasonography, the encapsulation of gaseous PFCs into polymer microsystems indeed improves their stability as compared to lipid/protein-stabilized PFC microbubbles. For vascular imaging, microsystems should be preferred as they provide more signal than nanosystems, although size should be optimized to allow intravenous injection. For extravascular imaging, for example tumor imaging, nanosystems should be preferred however there is plenty of room for improvement since

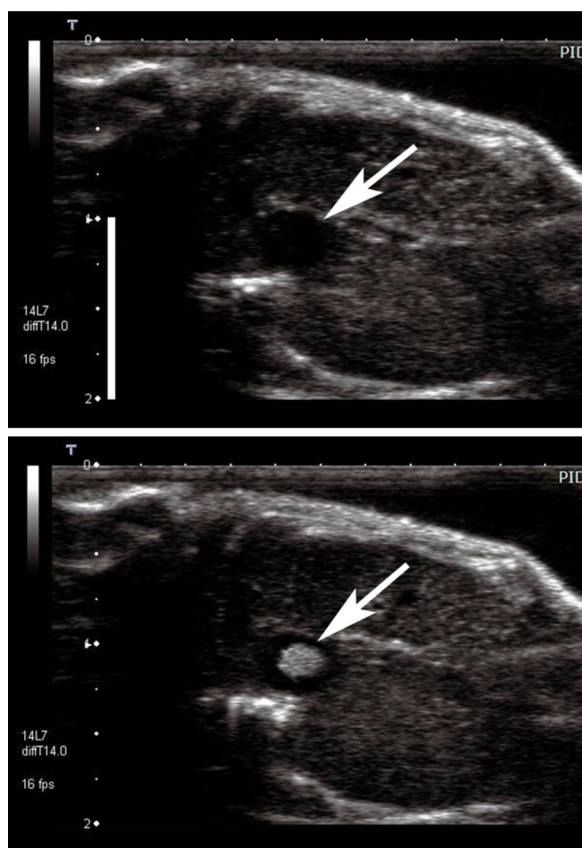


Fig. 6. Ultrasound images of mouse liver obtained in THI mode before (top) and after a 200- μL injection of a 50 mg/mL suspension of 150 nm-diameter PLGA-PFOB capsules ($T/R = 0.35$). The inferior vena cava (noted by an arrow) appears dark in the center of the top image, and presents significant enhancement after capsule injection (bottom). The scale bar represents 1 cm (from Ref. [14]).

their echogenicity is limited. The choice of polymer material, the shell thickness and the ultrasound frequency are worth investigating to improve these nanometric ultrasound contrast agents [65] as well as the liquid-to-gas transiting nanosystems [61,62].

3. PFC-loaded systems for ^{19}F -MRI

In addition to their interest for ultrasonography, PFCs may be used as contrast agents for fluorine magnetic resonance imaging (^{19}F -MRI). Conventional ^1H -MRI is based upon the property of hydrogen nuclei to align with and precess around an applied static magnetic field. The precessing hydrogen nuclei can be perturbed by radio frequency pulses, and the processes through which they return to their original aligned state can be exploited to give an image. The contrast comes from local differences in spin relaxation kinetics along the longitudinal (spin–lattice relaxation time T_1) and transverse (spin–spin relaxation time T_2) planes of the main magnetic field applied. All tissues have different T_1 and T_2 relaxation times based on their proton/water content. By varying the time to detect the signal (TE—time to echo) and the time to apply the next radiofrequency pulse (TR—time to repetition) one can optimize the signal. The benefits of fluorine in MRI compared to proton-based imaging are numerous. ^{19}F has a spin 1/2 nucleus and a gyromagnetic ratio very close to the proton allowing the use of proton spectrometers to detect fluorine. In addition, since ^{19}F is found only in trace amounts in biological tissues, there is no endogenous background signal. However, some technical problems of fluorinated compounds have slow down the use of ^{19}F MRI *in vivo*. Fluorinated molecules possess long T_1 and short T_2 relaxation times, resulting in long image acquisition times and MR signal attenuation, respectively. Moreover, the chemical shifts of ^{19}F NMR signals lead to chemical shift artifacts in the MR image [66]. However, using high fluorine concentrations such as those accessible with PFCs, the signal-to-noise ratio will be enhanced and the scanning time decreased for high-resolution images acquisition. One should note that ^{19}F -MRI is always combined with ^1H -MRI for anatomical superposition of the fluorine images. In addition, chemical shifts artifacts can be avoided by using specific ^{19}F -MRI sequences such as the multi spin echo (MSE) sequence developed for PFOB [67,68].

Lanza and Wickline have performed pioneering research on the potential of nanoemulsions for ^{19}F MRI. Nanoemulsions of PFOB or perfluoro-15-crown-5-ether (PFCE) of ~ 250 nm were formulated [69–71] with a ^{19}F concentration of approximately 100 M within each droplet, sufficient to yield a good ^{19}F -MRI signal [70]. Since they were targeted, nanoemulsion specific binding and local accumulation allowed to overcome the limited concentrations at the pathological site and the possible weakness of the fluorine signal as compared with proton. Following these results, untargeted nanoemulsions of PFCE were used as contrast agent for angiography with cardiovascular magnetic resonance at clinical strength [72]. The high level of ^{19}F signal was sufficient for angiography of small vessels *in vitro*, *ex vivo* on pig hearts for detecting coronary vessels and *in vivo* on rabbits after injection into carotid arteries. Nanoemulsions were also as beacons for tracking cells involved in pathological processes [73] or for molecular imaging of angiogenesis in xenograft mouse with peptides [74].

For ^{19}F -MRI, PFCE [15] or PERFECTA [75] are of particular interest since they exhibit a single ^{19}F -NMR resonance peak due to their chemical structure (Table 1). However these compounds were not extensively investigated from a toxicological point of view as it was the case for PFOB. Ahrens et al. [15] have developed PFCE nanoemulsions of 100–200 nm to label dendritic cells (DC) and track them after *i.v.* injection to mice using ^{19}F -MRI at a magnetic field of 11.7T (Fig. 7). Diabetogenic T cells migration to the pancreas after injection to mice was quantified by this method [16] and several other cell tracking applications have emerged [76]. Currently, a US company called Cellsense is developing all these cell tracking applications and clinical trials are in progress.

Beside nanoemulsions, polymer systems have been developed such as 20–30 nm micelles [77,78] but these need to be swollen with organic solvent to provide mobility to fluorinated chains, rendering their *in vivo* use unlikely. More promising are core-shell nanocapsules or liquid PFCs. Indeed, the very same PEGylated polyester nanocapsules used as ultrasound contrast agents revealed interesting for ^{19}F MRI and their long circulation properties as well as their ability to diagnose tumors were demonstrated *in vivo* on CT-26 xenograft tumors in mice [17,79] (Fig. 8). Similarly, Srinivas and co-workers entrapped different PFCs in PLGA nanoparticles with the aim of imaging primary

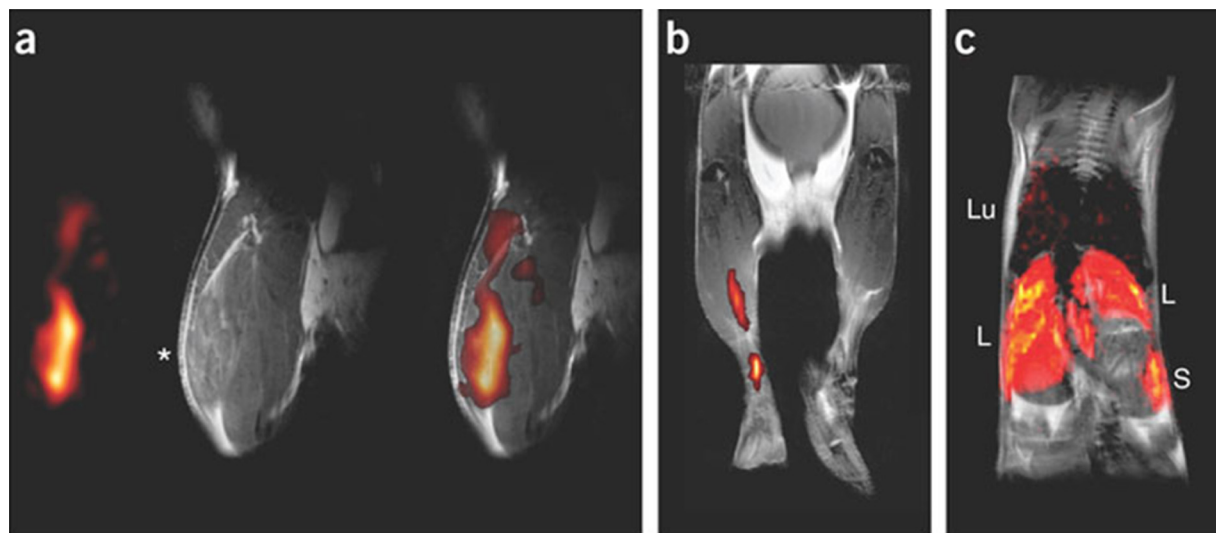


Fig. 7. *In vivo* MRI of PFPE-labeled dendritic cells (DCs) in mouse. The ^{19}F intensity is displayed on a “hot-iron” intensity scale and the ^1H image is shown in gray scale. (a) Mouse quadriceps after intramuscular injection of PFPE-labeled DCs, asterisk indicates injection site. Shown (from left to right) are ^{19}F and ^1H images and a superimposed $^{19}\text{F}/^1\text{H}$ image. (b) Composite image of DC migration into the popliteal lymph node following a hind foot pad injection. (c) Superimposed $^{19}\text{F}/^1\text{H}$ image through the torso following *i.v.* inoculation with PFPE-labeled DCs. Cells are apparent in the liver (L), spleen (S) and weakly in the lungs (Lu) (from Ref. [15]).

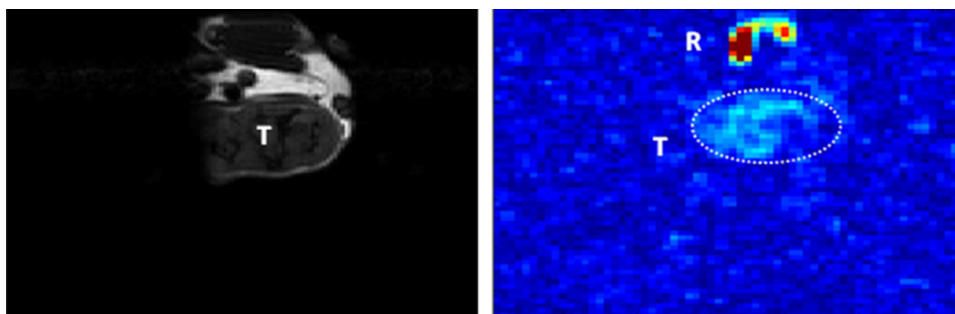


Fig. 8. Left: ^1H MRI image of a typical mouse bearing a CT-26.WT tumor (T) to yield anatomical information. Right: ^{19}F MRI image of the same mouse, the tumor is circled in white. R = ^{19}F reference to calculate nanocapsule tumor concentration (from Ref. [60]).

human dendritic cells [80]. The use of the emulsion technique and PVA as stabilizer induced the formation of nanosystems with a mean diameter higher than 200 nm that showed no toxic effects at a concentration of 20 mg per million cells and allowed to concentrate the PFC inside the cytosolic environment permitting also MR visualization after *in vivo* administration. By contrast with nanoemulsions, polyester nanocapsules can incorporate drugs in their polymer shell offering the possibility to become theranostic agents.

Overall, one can notice that studies on PFCs as ^{19}F MRI contrast agents are less numerous than studies on their use as ultrasound contrast agents. Indeed there is a real difficulty to access ^{19}F MRI set-ups since RF coils must be modified and MRI set-ups are less numerous than ultrasound echographs (an echograph is about 10 times less expensive than a MRI set-up). Significant improvements of ^{19}F -MRI contrast agents should translate into clinical practice thanks to specific ^{19}F -MRI sequences [67,68] eliminating chemical shifts artifacts, or to novel non-toxic fluorinated compounds exhibiting a single ^{19}F -NMR resonance peak [15,75].

4. Conclusion

Perfluorocarbons are of interest as contrast agents for both ultrasonography and fluorine magnetic resonance imaging. For ultrasonography, the most promising applications arise from gaseous PFCs encapsulated into microbubbles, although researchers are quickly progressing on optimizing liquid PFCs systems. In particular, an optimized nanosystem would allow to image ultrasound contrast agents beyond the vascular bed. For fluorine MRI, liquid PFCs have been formulated either as nanoemulsions or polymer nanocapsules and were successfully applied to track cells *in vivo*, to detect solid tumors or to perform molecular imaging. Clinical trials are in progress and the trend is to combine drugs with these contrast agents to yield theranostic systems.

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